Student: Xirong Zhang **Date:** 12/01/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 2.2

1. Find the inverse of the matrix.

$$\begin{bmatrix} 7 & 5 \\ 8 & 9 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. The inverse matrix is $\begin{bmatrix} \frac{9}{23} & -\frac{5}{23} \\ -\frac{8}{23} & \frac{7}{23} \end{bmatrix}$.

(Type an integer or simplified fraction for each matrix element.)

B. The matrix is not invertible.

2. Find the inverse of the matrix.

$$\begin{bmatrix} 5 & -4 \\ 3 & -2 \end{bmatrix}$$

Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. The inverse matrix is $\begin{bmatrix} -1 & 2 \\ -\frac{3}{2} & \frac{5}{2} \end{bmatrix}$.

(Type an integer or simplified fraction for each matrix element.)

B. The matrix is not invertible.

3. Determine whether the statement below is true or false. Justify the answer.

In order for a matrix B to be the inverse of A, both equations $AB = I$ and $BA = I$ must be true.

Choose the correct answer below.

A. The statement is true. The product of a matrix and its inverse is the identity matrix.

B. The statement is false. If $AB = I$ and $BC = I$, then A is one inverse of B and C is possibly another inverse of B.

C. The statement is false. It's possible that the product AB is defined and equals I, yet the product BA is not defined.

D. The statement is true. Since $AB = BA$, $AB = I$ if and only if $BA = I$.

4. Determine whether the statement below is true or false. Justify the answer.

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ab - cd \neq 0$, then A is invertible.

Choose the correct answer below.

- **A.** The statement is false. If $ad - bc \neq 0$, then A is invertible.
- **B.** The statement is false. If A is invertible, then $ab = cd$.
- **C.** The statement is true. The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $a \neq c$ and $b \neq d$.
- **D.** The statement is true. The inverse of A is $A^{-1} = \frac{1}{ab - cd}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. This expression is always defined when $ab - cd \neq 0$.

5. Determine whether the statement below is true or false. Justify the answer.

If A is an invertible $n \times n$ matrix, then the equation $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b}$ in $\mathbb{R}^n$.

Choose the correct answer below.

- **A.** The statement is true. Since A is invertible, $A^{-1}\mathbf{b}$ exists for all $\mathbf{b}$ in $\mathbb{R}^n$. Define $\mathbf{x} = A^{-1}\mathbf{b}$. Then $A\mathbf{x} = \mathbf{b}$.
- **B.** The statement is false. The matrix A is invertible if and only if A is row equivalent to the identity matrix, and not every matrix A satisfying $A\mathbf{x} = \mathbf{b}$ is row equivalent to the identity matrix.
- **C.** The statement is true. Since A is invertible, $A^{-1}\mathbf{b} = \mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$. Multiply both sides by A and the result is $A\mathbf{x} = \mathbf{b}$.
- **D.** The statement is false. The matrix A satisfies $A\mathbf{x} = \mathbf{b}$ if and only if A is row equivalent to the identity matrix, and not every matrix that is row equivalent to the identity matrix is invertible.

6. Determine whether the statement below is true or false. Justify the answer.

If A can be row reduced to the identity matrix, then A must be invertible.

Choose the correct answer below.

- **A.** The statement is true. Since A can be row reduced to the identity matrix, I is the inverse of A. Therefore, A is invertible.
- **B.** The statement is true. Since A can be row reduced to the identity matrix, A is row equivalent to the identity matrix. Since every matrix that is row equivalent to the identity is invertible, A is invertible.
- **C.** The statement is false. Since the identity matrix is not invertible, A is not invertible either.
- **D.** The statement is false. Not every matrix that is row equivalent to the identity matrix is invertible.

7. Solve the equation $AB = BC$ for A, assuming that A, B, and C are square matrices and B is invertible.

$A =$ BCB^{-1} (Simplify your answer.)

8. If A, B, and C are $n \times n$ invertible matrices, does the equation $C^{-1}(A + X)B^{-1} = I_n$ have a solution X? If so, find it.

Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** The solution is $X =$ $CB - A$.
- **B.** There is no solution.

9. Suppose A is $n \times n$ and the equation $Ax = b$ has a solution for each b in $\mathbb{R}^n$. Explain why A must be invertible. [Hint: Is A row equivalent to I_n?]

Choose the correct answer below.

- **A.** If the equation $Ax = b$ has a solution for each b in $\mathbb{R}^n$, then A has a pivot position in each row. Since A is square, the pivots must be on the diagonal of A. It follows that A is I_n. Therefore, A is invertible.
- **B.** If the equation $Ax = b$ has a solution for each b in $\mathbb{R}^n$, then A has a pivot position in each row. Since A is square, the pivots must be on the diagonal of A. It follows that A is row equivalent to I_n. Therefore, A is invertible.
- **C.** If the equation $Ax = b$ has a solution for each b in $\mathbb{R}^n$, then A does not have a pivot position in each row. Since A is square, and I_n is square, A is row equivalent to I_n. Therefore, A is invertible.
- **D.** If the equation $Ax = b$ has a solution for each b in $\mathbb{R}^n$, then A has one pivot position. It follows that A is row equivalent to I_n. Therefore, A is invertible.

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10. Find the inverse of the matrix, if it exists. Use the algorithm for finding A^{-1} by row reducing $[A \;\; I]$.

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 7 \end{bmatrix}$$

Set up the matrix $[A \;\; I]$.

$$[A \;\; I] = \begin{bmatrix} 2 & 2 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix}$$

(Type an integer or simplified fraction for each matrix element.)

Find the inverse. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

- **A.** The inverse matrix is $A^{-1} = \begin{bmatrix} \frac{7}{10} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{5} \end{bmatrix}$.

  (Type an integer or simplified fraction for each matrix element.)
- **B.** The matrix is not invertible.

11. Find the inverse of the given matrix, if it exists. Use the algorithm for finding A^{-1} by row reducing $[A\ I]$.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 1 & 4 \\ -3 & -4 & 4 \end{bmatrix}$$

Set up the matrix $[A\ I]$.

$$[A\ I] = \begin{bmatrix} 1 & 0 & 3 & 1 & 0 & 0 \\ -2 & 1 & 4 & 0 & 1 & 0 \\ -3 & -4 & 4 & 0 & 0 & 1 \end{bmatrix}$$

(Type an integer or simplified fraction for each matrix element.)

Find the inverse. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. The inverse matrix is $A^{-1} = \begin{bmatrix} \frac{20}{53} & \frac{-12}{53} & \frac{-3}{53} \\ \frac{-4}{53} & \frac{13}{53} & -\frac{10}{53} \\ \frac{11}{53} & \frac{4}{53} & \frac{1}{53} \end{bmatrix}$.

B. The matrix A does not have an inverse.

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12. Find the inverse of the given matrix, if it exists.

$$A = \begin{bmatrix} 1 & -2 & 1 \\ -4 & 9 & -5 \\ 2 & -6 & 4 \end{bmatrix}$$

Find the inverse. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

A. $A^{-1} =$ _____

(Type an integer or decimal for each matrix element.)

B. The matrix A does not have an inverse.